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Recursion – Homework

Advanced Programming

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Q1: Recursive String Compression

- Write a recursive function to compress a string by counting consecutive characters.

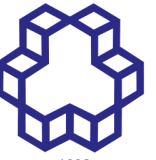
```
def compress_string_recur(s: str) -> str:
```

```
...
```

Examples :

```
compress_string_recur("aaabbc") → "a3b2c1"
```

```
compress_string_recur("abcd") → "a1b1c1d1"
```



Q2 : Deep Sum of Nested Lists

- Write a recursive function to sum of all integers inside a nested list of arbitrary depth.

```
def deep_sum(data):
```

```
...
```

Examples:

```
deep_sum([1, [2, [3, 4], 5], 6]) → 21
```

```
deep_sum([[[1]], 2, [[3, [4]]]]) → 10
```

Q3: Maximum Depth of Nested Lists

- Compute **maximum depth** of a nested list:

```
def deep_max_depth(data):
```

Example:

```
deep_max_depth([1, [2, [3, [4]]]]) → 4
```

Debugging Recursive Functions (Fix the bugs)

- find and fix the bugs in the following recursive function:

```
def flatten(lst):  
    if not lst:  
        return []  
    if isinstance(lst[0], list):  
        return flatten(lst[0])  
    else:  
        return [lst[0]] + flatten(lst[1:])
```